TC Cure Engine OussamaLabs

CURE RECORD
TOURING 205/55 R16 · demo serial D-24117
Tread TR-TOUR · 2026-09-28

TEST IDENTITY

TYRE	TREAD COMPOUND	SCAN INTERVAL	CHANNELS
TOURING 205/55 R16 · demo serial D-24117	TR-TOUR	5s	22 of 22

CURE START	PRESS OPEN	DEFLATION	EA SOURCE
240s Manual input (240s)	840s Manual input (600s)	850s Manual input (610s)	Compound DNA v1

ANALYSED BY	TAT FLOOR (PRIMARY)
demo.engineer R&D admin	ECM ≥ t90@150×1

CURE VERDICT — WINDOW-AWARE

FAIL **2 position(s) require action, 2 marginal.**
Window-aware: **18 PASS** · **2 MARGINAL** · **2 FAIL. 2 under-cured.**

ENGINEERING — ROOT-CAUSE BY ZONE

AP-01 — HARDWARE (asymmetry). Inadequate cure coincides with ΔT 16.9°C on AIT/AIB, A/BT/A/BB (whole zone). Check steam-line / bladder / platen on the cold side before the recipe.

BC-01 — WITHIN TOLERANCE (ran hot). 2 position(s) over the fixed cap but inside BC-01's reversion window (highest reversion index 23% of the 10 %-loss point). No cure action; if recurring, review setpoint margin.

⚐ BC-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

IL-01 — NO ACTION. All positions cured within spec.

RC-01 — NO ACTION. All positions cured within spec.

⚐ RC-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

SW-01 — NO ACTION. All positions cured within spec.

⚐ SW-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

TB-01 — NO ACTION. All positions cured within spec.

⚐ TB-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

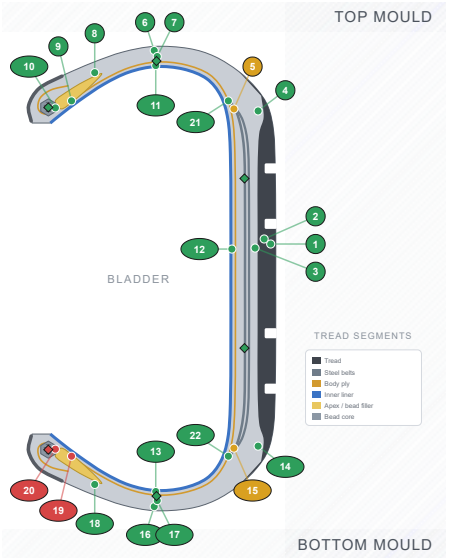
TR-TOUR — NO ACTION. All positions cured within spec.

⚐ TR-TOUR TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

UB-01 — NO ACTION. All positions cured within spec.

⚐ UB-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

SECTION MAP



PASS · MARGINAL · FAIL · excluded (fault)
not tested · estimated layer (no thermocouple)

PRESS HEALTH — TOP/BOTTOM Δ @ DEFLATION (ADVISORY)

ZONE	COMPOUND	TOP°C	BOT°C	ΔT	STATUS
Tread Shoulder	TR-TOUR	171.1	170.8	0.3	PASS
Belt Edge	BC-01	172.1	171.7	0.4	PASS
Sidewall Upper	SW-01	170.9	167.4	3.5	WARNING
Sidewall Middle	SW-01	172.5	167.6	4.8	WARNING
Mud Stopper	RC-01	169	161.5	7.4	WARN-HIGH
Apex Inside	AP-01	170.6	155.7	14.9	WARN-HIGH
Apex/Bead	AP-01	168.3	151.4	16.9	WARN-HIGH
Inner Liner	IL-01	173.1	165.6	7.5	WARN-HIGH
Under Belt	UB-01	174.5	174.3	0.3	PASS

CEQ DETAILED RESULTS

CH	POSITION	TMAX	CEQ@PO	CEQ@DEF	%CEF	CEQ	ASSESS
1	TCT TR-TOUR	172.6	17.86	18.49	138.1	PASS	PASS
2	TCM TR-TOUR	173.3	15.66	16.32	121.1	PASS	PASS
3	TCB TB-01	174.6	15.7	16.42	155.1	PASS	PASS
4	TST TR-TOUR	171.4	12.65	13.24	97.9	PASS	PASS
5	SB2T BC-01	172.6	12.21	12.94	83.9	FAIL	MARGINAL
6	SWTMT SW-01	171.1	21.79	22.41	156.6	PASS	PASS
7	SWUMT SW-01	172.6	21.77	22.45	156.5	PASS	PASS
8	SWULT RC-01	169	13.63	14.19	110	PASS	PASS
9	AIT AP-01	170.7	12.43	13.05	116.1	PASS	PASS
10	A/BT AP-01	168.7	9.48	10.01	88.5	PASS	PASS
11	ILT IL-01	176.6	33.66	34.54	122	PASS	PASS
12	ILM IL-01	180.5	34.04	35.09	123.3	PASS	PASS
13	ILB IL-01	168.2	18.06	18.58	65.4	PASS	PASS
14	TSB TR-TOUR	171.2	12.45	13.03	96.3	PASS	PASS
15	SB2B BC-01	172.5	12	12.72	82.5	FAIL	MARGINAL
16	SWTMB SW-01	167.6	17.23	17.73	123.9	PASS	PASS
17	SWUMB SW-01	167.6	15.71	16.22	113	PASS	PASS
18	SWULB RC-01	161.8	8.42	8.77	67.9	PASS	PASS
19	AIB AP-01	156	4.38	4.62	40.9	FAIL	FAIL
20	A/BB AP-01	152.2	2.92	3.1	27.3	FAIL	FAIL
21	UBT UB-01	174.6	13.34	14.07	103	PASS	PASS
22	UBB UB-01	174.3	12.99	13.7	100.3	PASS	PASS

TAT — ECM ENVELOPE PER COMPOUND (PRIMARY) · RAW MIN≥150 SECONDARY

COMPOUND	ECM WORST	FLOOR	MARGIN	ECM MAX	TIER	RAW150 W/FLR	ENVELOPE
TR-TOUR MARCHING	27.74	12.88	+115.4%	33.08	NO-CEILING-DATA	9.17/13.52 ⚐	ok
TB-01 PLATEAU	36.42	9.95	+266%	36.42	OK	10.25/10.45 ⚐	ok
BC-01 PLATEAU	31.5	14.82	+112.5%	32.07	OK	9.83/15.56 ⚐	ok
SW-01 PLATEAU	22.92	13.8	+66.1%	32.79	OK	8.17/14.49 ⚐	ok
RC-01 PLATEAU	13.32	12.23	+8.9%	22.96	OK	5.42/12.84 ⚐	ok
AP-01 PLATEAU	7.84	10.6	-26.1%	26.4	OK	1.75/11.13	under-cure
IL-01 MARCHING	23.39	27.51	-15%	49.5	NO-CEILING-DATA	8.42/28.89	under-cure
UB-01 PLATEAU	35.68	12.71	+180.7%	36.6	OK	10.5/13.35 ⚐	ok

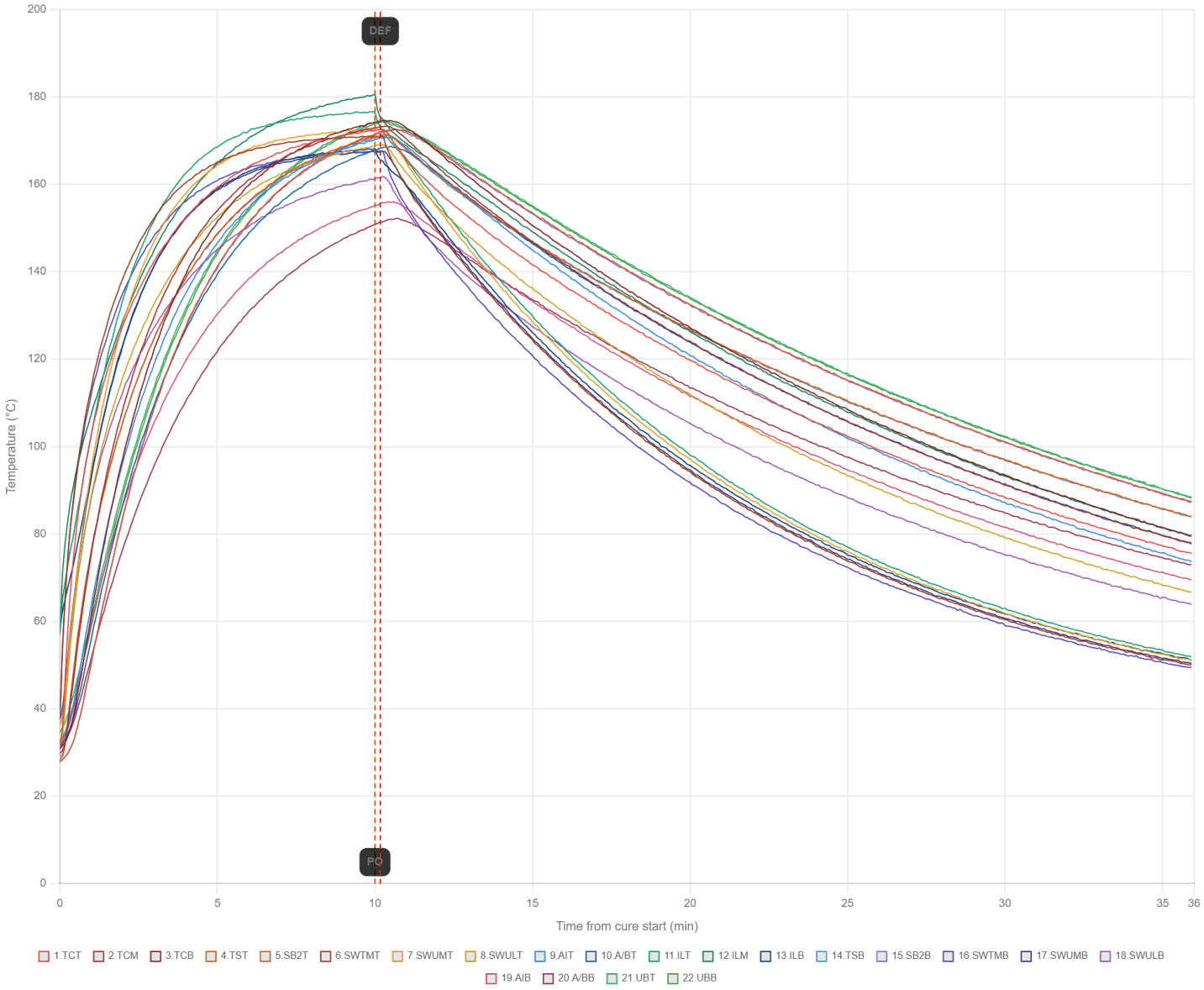
ESTIMATED LAYERS — BY CONSTRUCTION (INTERPOLATED BETWEEN BRACKETING TCS; NOT IN THE HEADLINE)

SPOT	COMPOUND · LAYER	TMAX	CEQ@PO	CEQ@DEF	%CEF	VERDICT
BSK-T	SK-01 · Belt skim (above the belts, under the tread base) — top side	173.6	13.57	14.28	114.6	PASS
BSK-B	SK-01 · Belt skim (above the belts, under the tread base) — bottom side	173.5	13.47	14.17	113.7	PASS
INS-T	IN-01 · Insulation (between inner liner and body ply) — top side	174.5	25.59	26.32	203	PASS
INS-B	IN-01 · Insulation (between inner liner and body ply) — bottom side	167.9	16.29	16.79	129.2	PASS
BWC-T	BW-01 · Bead wire coat (bead core, under the apex) — top side	168.7	10	10.59	39.8	PASS
BWC-B	BW-01 · Bead wire coat (bead core, under the apex) — bottom side	152.2	2.78	2.97	11.1	FAIL rev. 3%

POST-CURE ACCUMULATION (AFTER PRESS-OPEN)

CH	POSITION	CEQ@PO	CEQ@END	POST-CURE %	%CEF@PO	%CEF@END
1	TCT TR-TOUR	17.86	28.19	+58%	138.1	218.1
2	TCM TR-TOUR	15.66	29.12	+86%	121.1	225.2
3	TCB TB-01	15.7	32.11	+105%	155.1	317.3
4	TST TR-TOUR	12.65	25.58	+102%	97.9	197.9
5	SB2T BC-01	12.21	31.16	+155%	83.9	214.2
6	SWTMT SW-01	21.79	27.49	+26%	156.6	197.6
7	SWUMT SW-01	21.77	29.29	+35%	156.5	210.6
8	SWULT RC-01	13.63	21.58	+58%	110	174
9	AIT AP-01	12.43	24.76	+99%	116.1	231.1
10	A/BT AP-01	9.48	21.83	+130%	88.5	203.8
11	ILT IL-01	33.66	42.13	+25%	122	152.6
12	ILM IL-01	34.04	49.96	+47%	123.3	181
13	ILB IL-01	18.06	23.75	+32%	65.4	86.1
14	TSB TR-TOUR	12.45	25.18	+102%	96.3	194.8
15	SB2B BC-01	12	30.62	+155%	82.5	210.5
16	SWTMB SW-01	17.23	21.59	+25%	123.9	155.2
17	SWUMB SW-01	15.71	21.29	+36%	113	153.1
18	SWULB RC-01	8.42	12.92	+53%	67.9	104.2
19	AIB AP-01	4.38	9.53	+118%	40.9	89
20	A/BB AP-01	2.92	7.66	+162%	27.3	71.5
21	UBT UB-01	13.34	32.51	+144%	103	251
22	UBB UB-01	12.99	31.79	+145%	100.3	245.5

TEMPERATURE PROFILE



SIGN-OFF

Section cut performed: _____ All TC locations verified: _____

Tester: _____ Approved by: _____

Comments: _____